

BALALAIKA-LONGFORM: A RUSSIAN SPEECH CORPUS FOR CONTINUOUS LONG-FORM TEXT-TO-SPEECH

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ABSTRACT

Long-form text-to-speech must retain requested words over extended generations, yet sentence-level training and evaluation can hide omissions and early stops. We introduce Balalaika-Longform, a Russian corpus of 189 hours in continuous units of 30 seconds to 15 minutes. Paired long sequences and short windows support fine-tuning comparisons on the same source recordings. The accompanying RuLongTTS evaluation retains every synthesis attempt. Alongside an earlier controlled CosyVoice3 experiment, a separate completed evaluation uses 50 published-text pairs from 22 works and 21 held-out voices across three autoregressive backbones and F5-TTS flow matching. At approximately 1,200 words, CosyVoice3’s Long-minus-Short WER change is -52.7 percentage points. For the other backbones, paired intervals favor Long-SFT-Punct for Qwen3-TTS, VoxCPM2 and F5-TTS. These results characterize the effect of preserving training sequence length under a fixed continuous-generation protocol; they do not establish listener-rated quality or superiority to chunked synthesis.

Index Terms— text-to-speech, Russian speech, speech corpus, long-form synthesis

1. INTRODUCTION

Audiobooks, lectures and podcasts require speech synthesis to preserve content and voice beyond a sentence. Short outputs can conceal omissions, repetitions and early stops, and listeners judge sentences and paragraphs differently [1]. Existing pipelines can obtain long audio by splitting text into short generations [2, 3]. Studying one uninterrupted generation requires data and evaluation that retain longer context.

We introduce Balalaika-Longform for this purpose in Russian. The Balalaika pipeline [4] supplies context-preserving segmentation, transcripts, provenance and quality screening; the corpus retains continuous speech units up to 15 minutes. Paired short-window and long-sequence training views allow a source-controlled comparison of sequence length, with channel-disjoint train, development and test splits.

We evaluate these views on CosyVoice3, Qwen3-TTS, VoxCPM2 and F5-TTS. The earlier controlled CosyVoice3 experiment tests duration under closely matched training exposure (Table 2). A separate published-text comparison tests 50 fixed text-voice pairs from 22 works (Table 3), retaining every failure in word-error and correct-word-recall estimates. Paired Short/Long comparisons quantify the effect and its uncertainty for all four backbones. The paired intervals favor long targets across the four tested backbones.

Our contributions are the long-form corpus, paired training views and an all-attempt evaluation of corpus use across generation families. The experiments separate this data question from the choice to generate continuously or in chunks. Transcript-source

and formatting changes, and VoxCPM2’s retained-audio differences, remain explicit confounds rather than isolated duration effects.

2. RELATED WORK

Speech corpora for synthesis. Open TTS corpora make data quality and coverage reproducible research choices. LibriTTS-R improves the acoustic quality of an existing English corpus through speech restoration [5], while HiFiTTS-2 scales audiobook-derived data with quality filtering and detailed metadata [6]. For Russian, RUSLAN provides carefully recorded single-speaker speech [7], and Balalaika provides a pipeline for context-preserving segmentation, multi-ASR transcription, and speaker-purity screening [4]. We build on the latter to produce a corpus organized around long continuous speech units. Its paired short and long training views let us examine sequence duration while keeping the source recordings fixed.

Approaches to long-form generation. Recent work addresses long speech through inference and model design. MagpieTTS-LF carries state and text history across sentence chunks without retraining [8]. VoiceStar combines positional modeling and a training scheme for duration control and extrapolation [9]. F5-TTS offers a non-autoregressive flow-matching alternative [10]; its standard long-text inference splits the text and joins separately synthesized chunks [3]. These approaches show that long-form behavior can be improved through several mechanisms. Our study examines the contribution of training data: whether preserving long targets helps existing models under a fixed uninterrupted-inference protocol. It does not assume that long-form training is necessary for every synthesis method.

Backbones and evaluation. CosyVoice [11, 12], Qwen3-TTS [13], and VoxCPM2 [14] provide open autoregressive models with different speech representations and generation behavior; Qwen3-TTS also reports a dedicated long-speech evaluation. We use their public fine-tuning paths to assess the same corpus across backbones. LFS-Bench evaluates long-form speech across scenarios and dimensions including content, acoustics, and expressiveness [15]. Our benchmark addresses a narrower training-data question: it controls input duration, retains failed attempts in the reported metrics, and evaluates whether a single generation reads the requested text. Speaker similarity and quality proxies are reported separately because completion alone does not establish perceptual quality.

3. CORPUS AND EVALUATION DESIGN

3.1. Corpus, data splits, and training arms

Balalaika-Longform is released openly on Hugging Face.

[PENDING: Replace `\hfdataseturl` with the live repository URL.]

The open-source Balalaika pipeline [4] provided context-aware segmentation, multi-ASR ROVER transcription, and quality and

speaker-purity screening. DeepFilterNet3 [16] attenuated applause and background noise on working copies; originals were retained unchanged and only enhanced derivatives entered training.

Balalaika-Longform v3.1 contains 6,380 Creative-Commons Russian segments (189.13 h; 30–900 s, single speaker) from 313 videos and 41 channels, with text from recognizer-output voting (ROVER) [17] over available ASR hypotheses. We partition by YouTube channel before windowing: all segments of a channel belong to one split (Table 1). All SFT uses train only; dev supplies validation and the two-voice evaluation’s text selection. Channel identity is a conservative document/speaker proxy, supplemented by exact-text and 8-gram leakage checks and a speaker-embedding gate across splits.

Table 1. Corpus splits before training-text exclusions; 24 additional held-out segments (0.88 h) fail quality gates.

Split	Segments	Hours	Videos
Train	6,043	170.68	265
Dev	120	8.50	17
Test	193	9.07	30

Training starts from 6,043 long units or 28,532 short windows of the same recordings (170.68 h, 1,257,944 words before empty-text exclusions). For CosyVoice3, each long speech-token sequence concatenates its windows. Short-SFT and Long-SFT use ROVER text; Long-SFT-Punct keeps the Long-SFT audio tokens but substitutes punctuated, capitalized GigaAM-v3-e2e-CTC transcripts. This also changes words: after the same evaluation normalization, 95.0% of the 6,041 training transcript pairs differ, with 5.4% pooled word-edit disagreement relative to ROVER. This is not a punctuation-only ablation.

3.2. Fine-tuning to assess corpus utility

The main comparison changes training-sequence length while controlling exposure to the corpus. All CosyVoice3 arms start from the same checkpoint and fine-tune only the language model; the speech tokenizer, acoustic generator, and speaker encoder stay fixed. They use 3,000 updates with the same learning rate and loss normalization. Relative to Long-SFT, observed target-token exposure differs by 1.4% for Short-SFT and by 1.27% for Long-SFT-Punct.

We fine-tune Qwen3-TTS and VoxCPM2 with their public training code [18, 19], using punctuated text in both the short and long arms. Within-backbone exposure differs by 0.17% for Qwen3-TTS and 0.3% for VoxCPM2. VoxCPM2’s context-length filter excludes 144 long units lasting 517–899 s, or 18.2% of the audio, leaving units of at most 661 s. The short arm retains windows from these recordings, so its comparison does not fully isolate duration from corpus composition. The CosyVoice3 and Qwen3-TTS arms share recording sets.

For F5-TTS, both arms fine-tune the full Russian v1 Base model with the flow-matching objective and punctuated transcripts. They traverse the same 6,038 recordings in the same order for three passes (18,114 updates), using intact recordings or short windows. Long/Short consume 510.92/510.26 hours of training audio, within 0.24% of the corresponding Qwen replication budgets; final exponential-moving-average (EMA) weights are evaluated without score-based checkpoint selection. Both arms exclude records lacking usable windows; residual normalized transcript disagreement is 0.04%. Full training and runtime settings accompany the code.

3.3. State-continuous inference and evaluation sets

Continuous inference uses one target text and one voice-conditioning event, followed by an unsplit autoregressive trajectory or full-sequence flow solve. External splitting, state resets, best-of- N selection and regeneration are disabled; all attempts are retained. The CosyVoice3 text splitter is bypassed. Operational context, output and acoustic-buffer limits are fixed across checkpoints within each backbone and documented with the code. F5-TTS estimates duration from prompt audio and text length; it has no EOS decision, and planned duration is not a learned stopping measure.

The earlier two-voice controlled set combines six source texts (four development recordings and two constructed stress texts), five nested duration ranges and two held-out voices: 60 attempts per arm (Table 2). The new published-text set uses 50 pairs from 22 works and 21 reference speakers held out from fine-tuning: eight new external readers and 13 used in the earlier evaluation. Each work supplies one long passage near 1,200 normalized words; 14 short and 14 medium prefixes target 75 and 300 words. Length ranges are 60–90, 240–360 and 960–1,440 words, rather than reference-audio durations. Inputs are literal, sentence-complete published spans, screened against local training and prior evaluation text. The same fixed pairs are used by every arm.

A submission-deadline reduction from 120 to 50 pairs followed inspection of preliminary Base/Short results and Long-run progress. The deterministic selector used only source and speaker metadata, without scores or output duration; this scope amendment is not a pre-synthesis selection. The 120-pair study is deferred. The completed comparison includes all 13 continuous conditions. All input is punctuated and capitalized. Among the CosyVoice3 arms, only Long-SFT-Punct matches this format in training; the two adapted arms of each other backbone are format-matched.

3.4. Evaluator and paired uncertainty

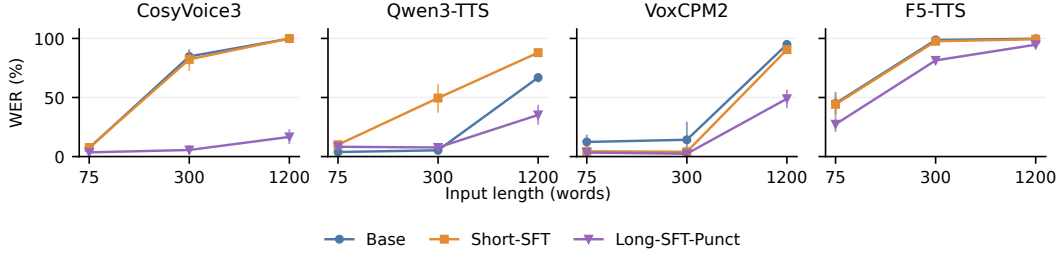
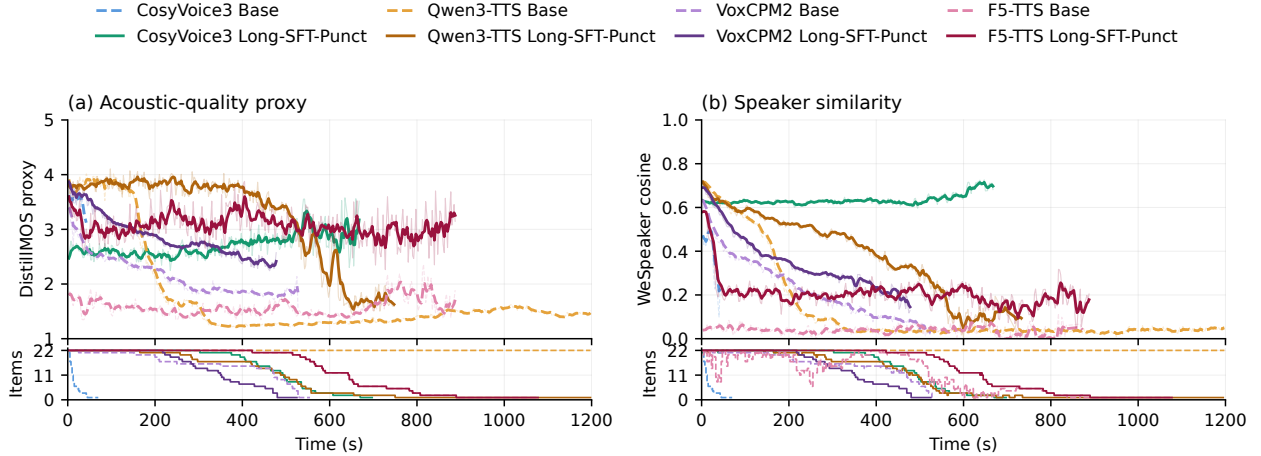
GigaAM-v3-RNNT [20] transcribes every generated output. Mean per-attempt WER counts substitutions, deletions and insertions relative to reference words; correct-word recall is aligned correct words divided by reference words. Unusable audio becomes an empty hypothesis. Failed and empty outputs remain in every denominator, and WER is not clipped at 100%. The published-text set has no paired human recording, so no human-ASR floor or floor-dependent completion label is used for its quality results.

Table 2 retains the earlier completion rule, frozen before SFT: an autoregressive EOS before 95% aligned text coverage is early; remaining outputs are degraded by WER at least 30 points above a human-ASR floor or fixed repetition criteria. Watchdog stops retain their label. Constructed texts lack human recordings and skip the WER-floor test, so complete does not imply a faithful transcript. Earlier F5 missing-tail events are incomplete text, never EOS.

For the new set, each backbone’s primary contrast compares its long and short arms (Long-SFT for CosyVoice3, Long-SFT-Punct for the other backbones) on the 22 longest inputs. Paired pointwise 95% intervals use 10,000 resamples of the 22 work clusters, retaining associated prefixes and voices; independently resampling work and voice levels is a crossed-factor sensitivity check. Intervals condition on fitted checkpoints and inference seed, omit retraining variation, and do not adjust for multiple comparisons. Earlier two-voice intervals resample its six source texts separately.

Table 2. CosyVoice3 results on the two-voice set by input duration; all synthesis attempts are retained.

Bucket	Target duration	Complete $\uparrow$ (%)				WER-all $\downarrow$ (%)			
		Base	Short-SFT	Long-SFT	Long-SFT-Punct	Base	Short-SFT	Long-SFT	Long-SFT-Punct
B0	20–40 s	83.3	91.7	100.0	100.0	7.18	7.07	3.45	2.93
B1	60–90 s	58.3	41.7	83.3	75.0	14.56	17.04	11.70	9.61
B2	2–3 min	0.0	0.0	100.0	100.0	90.86	82.64	2.74	3.42
B3	4–6 min	0.0	0.0	41.7	75.0	98.07	97.77	47.22	36.66
B4	8–12 min	0.0	0.0	33.3	66.7	99.82	99.89	37.22	19.15
All	20 s–12 min	28.3	26.7	71.7	83.3	62.10	60.88	20.47	14.36

**Fig. 1.** All-attempt WER for continuous synthesis near 75, 300 and 1,200 input words (14, 14 and 22 pairs per condition); bars show pointwise work-cluster 95% intervals. Long-SFT-Punct denotes long-target training on punctuated transcripts. The separate CosyVoice3 Long-SFT arm is reported in Table 3. Qwen3-TTS, VoxCPM2 and F5-TTS also use punctuated text for Short-SFT.**Fig. 2.** Windowed diagnostics on the same 22 longest inputs per condition, with training labels as in Fig. 1. Base and Long-SFT-Punct are shown; all 13 conditions are retained in the companion figure. Both metrics use 5-s windows and a 2.5-s hop; speaker windows require at least 1 s of voiced audio. Faint curves are item-weighted means, bold curves exponential smoothing ($\alpha = 0.3$). Lower strips count contributing items; scores require at least two. Neither proxy measures text fidelity.

4. EXPERIMENTS

4.1. Earlier controlled duration comparison

Long supervision improves native CosyVoice3 reading with two reference voices (Table 2). Short-SFT resembles Base, whereas Long-SFT gains 45.00 completion points [33.33, 56.67] over Short-SFT under matched exposure. Neither Base nor Short-SFT completes an item beyond 90 s. The sentence-split control completes all items at 3.84% WER; our continuous protocol claims no superiority over it. Repeated decoding shifts individual bucket percentages by up to ~ 20 points, so these single-seed rows are descriptive.

Punctuated transcripts add a secondary gain. Long-SFT-Punct reaches 83.3% completion; the prespecified pooled analysis gives +10.62 completion points [4.55, 16.38] and -6.69 WER points over

Long-SFT. This comparison changes both transcript source and formatting. Giving Long-SFT lowercased, unpunctuated input matched to its training format yields 88.3% completion and 10.07% WER; Long-SFT-Punct does not improve significantly on that control. The gain is therefore consistent with format matching, while a separate benefit of punctuation content remains unestablished: this control uses only the two-voice set and one seed, and training and input formats were not fully crossed. Three preregistered mixtures with external short clips did not improve Long-SFT and increased repetition.

Table 3. Published-text evaluation at approximately 1,200 input words: 22 attempts from 22 works per condition. WER and correct-word recall include failed outputs. Brackets show work-cluster bootstrap 95% intervals for mean WER. The completed group comprises all 13 continuous conditions.

Backbone	Arm	WER ↓ (%) [95% CI]	Correct-word recall ↑ (%)
CosyVoice3	Base	99.8 [99.7, 99.9]	0.2
	Short-SFT	99.9 [99.9, 100.0]	0.1
	Long-SFT	47.3 [37.2, 57.7]	69.5
	Long-SFT-Punct	16.6 [10.5, 23.2]	85.9
Qwen3-TTS	Base	66.9 [62.8, 70.8]	34.4
	Short-SFT	88.0 [85.5, 90.3]	12.2
	Long-SFT-Punct	35.2 [27.0, 43.9]	67.1
VoxCPM2	Base	94.9 [91.0, 98.0]	5.2
	Short-SFT	90.5 [86.9, 93.9]	10.5
	Long-SFT-Punct	49.0 [40.8, 56.7]	54.9
F5-TTS	Base	99.7 [99.7, 99.8]	0.3
	Short-SFT	99.3 [99.2, 99.5]	0.7
	Long-SFT-Punct	94.7 [94.4, 95.0]	5.4

4.2. Content fidelity across backbones

The completed published-text comparison tests the training views on new source passages across four backbones (Table 3). For the three autoregressive models, Long-minus-Short WER changes at approximately 1,200 words are CosyVoice3: -52.7 pp $[-62.7, -42.2]$; Qwen3-TTS: -52.7 pp $[-61.7, -43.4]$; VoxCPM2: -41.5 pp $[-47.7, -35.4]$. Negative changes indicate fewer word errors. All are paired effects on the same 22 work passages; short and medium conditions in Fig. 1 show how absolute error changes with input length. These results do not treat the 50 dependent attempts as 50 independent source passages.

Correct-word recall complements WER by measuring how much reference content is actually recognized, without counting substitutions as correct. The table retains poor and failed readings rather than conditioning on completion. CosyVoice3’s punctuated-transcript arm is a separate format/source comparison, while VoxCPM2’s Long/Short comparison also changes retained corpus composition.

4.3. Flow matching and inference protocol

F5-TTS extends the paired comparison to a flow-matching model. Its longest-input Long-minus-Short WER change is -4.6 pp $[-4.9, -4.4]$. Table 3 also reports absolute WER and correct-word recall: a relative change alone does not establish reliable long-text reading. The duration schedule is shared across checkpoints, so this comparison evaluates content realization rather than learning an EOS decision.

Chunking provides a distinct practical reference. On the *earlier* recorded-text 53-pair set, F5 Base WER was 3.58% with sentence chunking and 73.35% with continuous synthesis. Those values use different texts from Table 3 and are not rows of the new comparison. Chunked controls are not included in the completed new-set group; the earlier result establishes no new-set ranking.

4.4. Evaluation checks and practical limits

Earlier secondary-ASR checks supported the CosyVoice3 ranking on recorded texts; the new-set WER uses one RNNT recognizer. Figure 2 adds windowed DistillMOS [21] and WeSpeaker [22] measurements on the same new outputs. F5 Long-SFT-Punct has higher Dis-

tillMOS than Base despite high WER; Qwen3-TTS and VoxCPM2 still show speaker drift. Curves condition on scorable windows, with support shown below. Truncated outputs can score well, so these proxies complement all-attempt WER and do not replace listening tests.

An earlier 5–10-minute CosyVoice3 audit found continuous generation used $3.67\times$ the wall time and $3.56\times$ the peak memory of sentence splitting; these costs are protocol-specific.

5. LIMITATIONS

The published-text set covers 22 selected works and repeated reference voices, not all Russian speech. Its deadline-driven scope amendment followed inspection of earlier run summaries, even though the subset selector ignored outcomes. The 120-pair evaluation and sealed 40-text benchmark remain incomplete. Earlier CosyVoice3 checkpoint selection and controlled evaluation share development data. Each backbone has one training seed per arm; paired intervals do not include retraining, repeated-decoding or selection uncertainty.

GigaAM contributes training transcripts and evaluates outputs, leaving possible recognizer bias. No listening study establishes naturalness or speaker preference. Unknown base-model pretraining overlap is not ruled out by local text screening. Punctuation and transcript-source effects are not fully isolated. VoxCPM2’s long arm retains 81.8% of the audio, so matched exposure cannot remove the composition confound; earlier speaker drift also remains unresolved. One F5 model does not establish architectural causes or generality to all non-autoregressive models.

6. CONCLUSION

Balalaika-Longform preserves long continuous Russian speech with paired short-window training views. The controlled CosyVoice3 study and the completed published-text comparison provide separate evidence about sequence length. Across CosyVoice3, Qwen3-TTS, VoxCPM2 and F5-TTS, the study reports absolute content fidelity and paired Short/Long changes. The paired intervals favor long targets across the four tested backbones. The corpus supports studying sustained reading while keeping content fidelity and perceptual quality distinct.

7. COMPLIANCE WITH ETHICAL STANDARDS

Balalaika-Longform is openly distributed through Hugging Face with per-source provenance and license metadata. Original recordings remain governed by their respective source licenses; the release does not relicense third-party material. The public repository documentation should retain source attribution and a takedown/contact procedure. **[PENDING: Confirm the final URL, attribution wording, and takedown contact.]**

8. REFERENCES

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